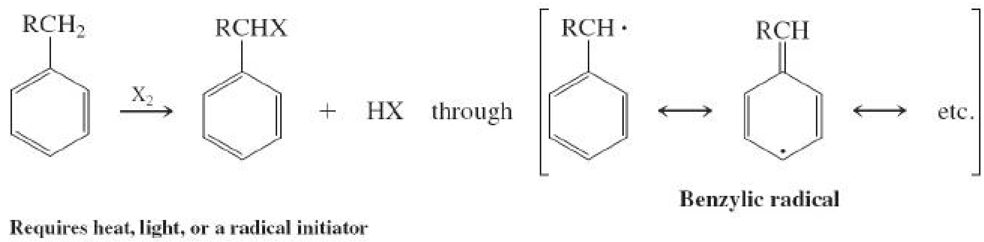
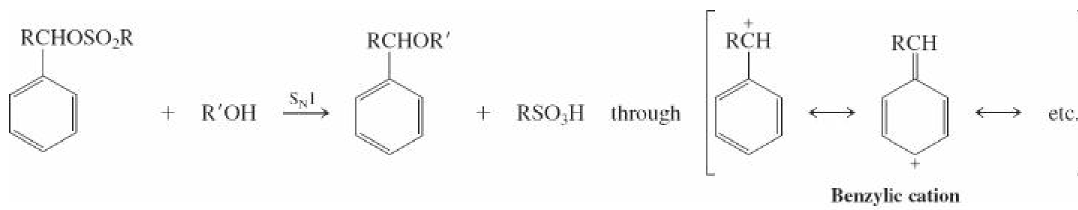
New Reactions

Benzylic Resonance

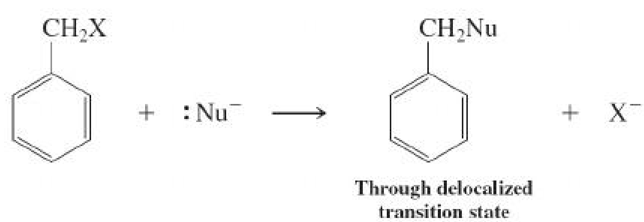
1. Radical Halogenation (Section 22-1)



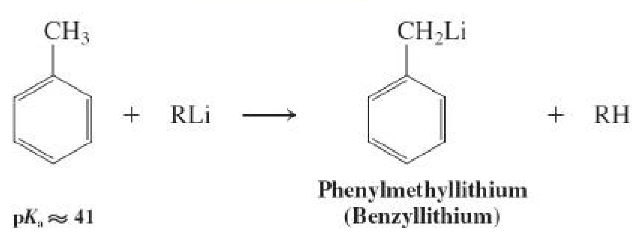
2. Solvolysis (Section 22-1)



3. SN2S_N2 Reactions of (Halomethyl)benzenes (Section 22-1)

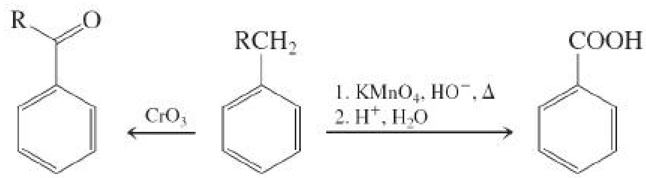


4. Benzylic Deprotonation (Section 22-1)

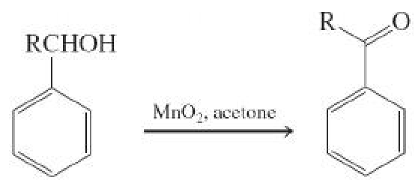


Oxidation and Reduction

5. Oxidation (Section 22-2)

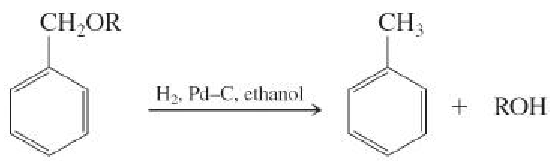


Benzylic alcohols



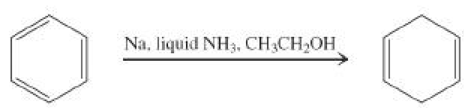
6. Reduction (Section 22-2)

Hydrogenolysis

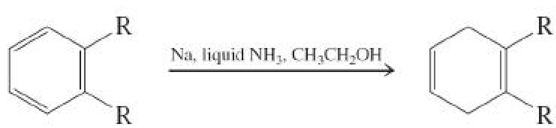


$C_6H_5CH_2$ is a protecting group for ROH.

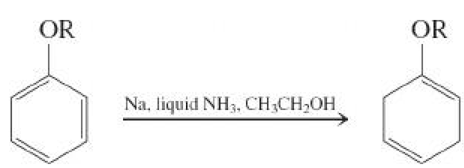
One-Electron (Birch) Reduction

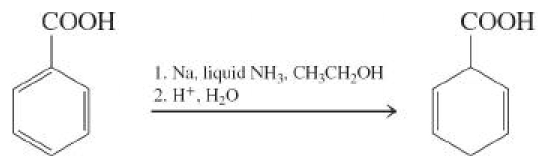


Regioselectivity:



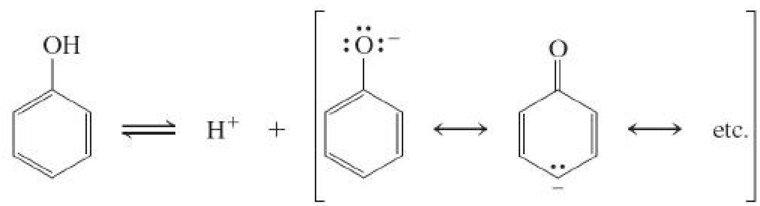
R = alkyl





Phenols and Ipso Substitution

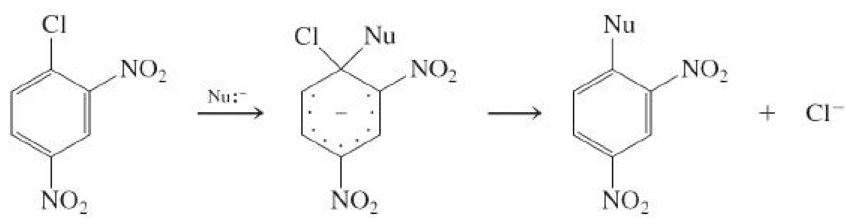
7. Acidity (Section 22-3)



p$K_a \approx 10$
Much stronger acid than simple alkanols.

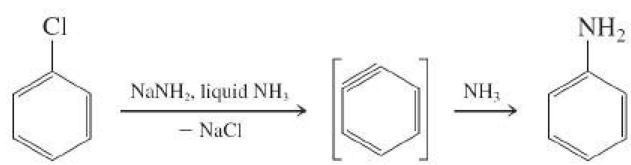
Phenoxide ion

8. Nucleophilic Aromatic Substitution (Section 22-4)

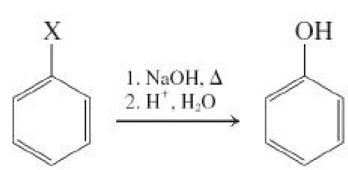


Nucleophile attacks at ipso position.

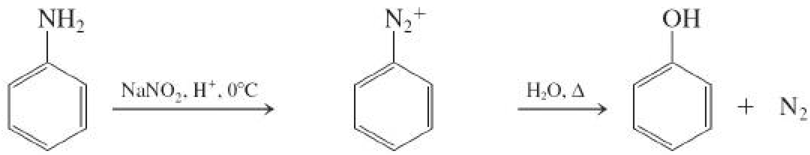
9. Aromatic Substitution Through Benzyne Intermediates (Section 22-4)



Nucleophile attacks at both ipso and ortho positions.



10. Arenediazonium Salt Hydrolysis (Section 22-4)

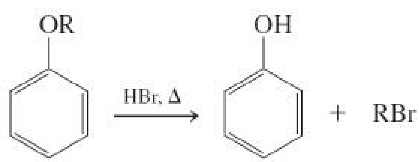


Benzenediazonium cation

11. Substitution with Pd Catalysis

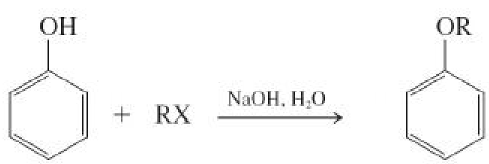
Reactions of Phenols and Alkoxybenzenes

12. Ether Cleavage (Section 22-5)



Aryl C–O bond is not cleaved.

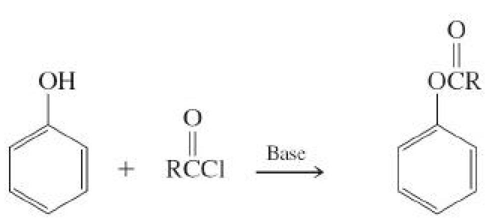
13. Ether Formation (Section 22-5)



Alkoxybenzene

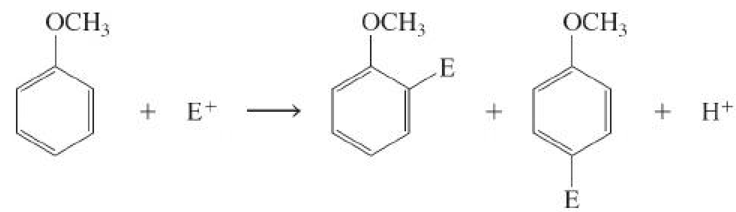
Williamson method (Section 9-6)

14. Esterification (Section 22-5)

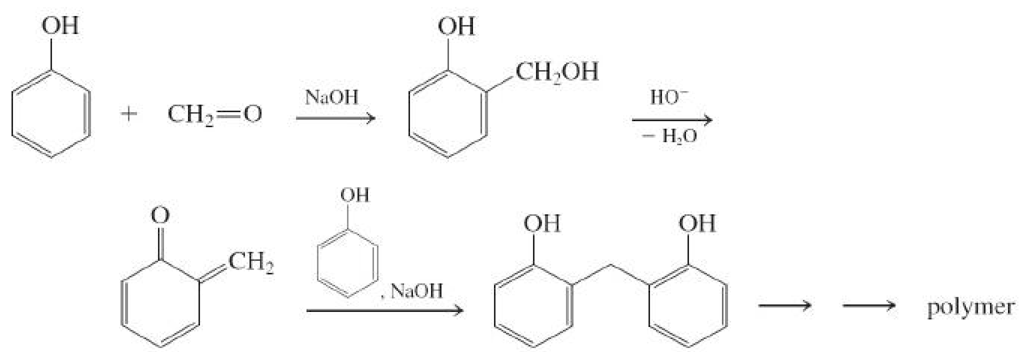


Phenyl alkanoate

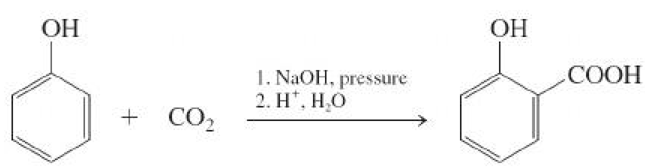
15. Electrophilic Aromatic Substitution ([Section 22-6](#))



16. Phenolic Resins ([Section 22-6](#))

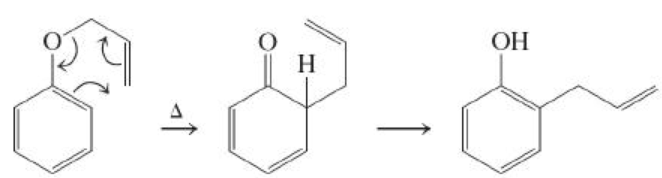


17. Kolbe Reaction ([Section 22-6](#))

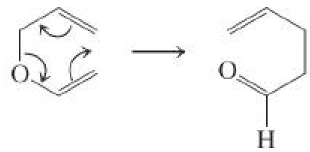


18. Claisen Rearrangement ([Section 22-7](#))

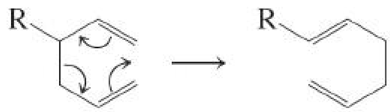
Aromatic Claisen rearrangement



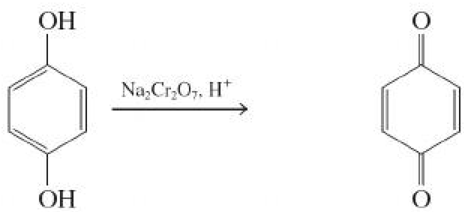
Aliphatic Claisen rearrangement



19. Cope Rearrangement ([Section 22-7])

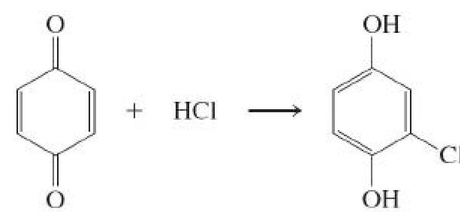


20. Oxidation ([Section 22-8])

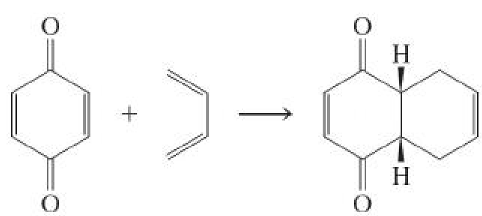


2,5-Cyclohexadiene-1,4-dione
(*p*-Benzoquinone)

21. Conjugate Additions to 2,5-Cyclohexadiene-1,4-diones (p*p*-Benzoquinones) ([Section 22-8])

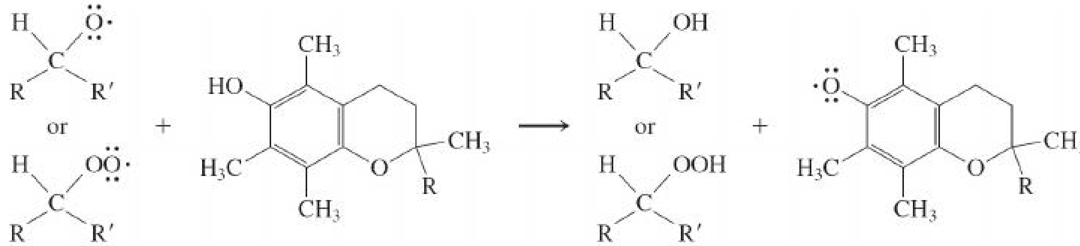


22. Diels-Alder Cycloadditions to 2,5-Cyclohexadiene-1,4-diones (p*p*-Benzoquinones) ([Section 22-8])



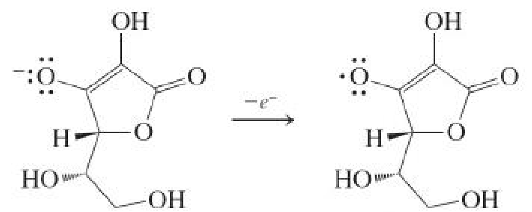
23. Lipid Peroxidation ([Section 22-9])

24. Inhibition by Antioxidants ([Section 22-9](#))



Vitamin E (or BHA or BHT)

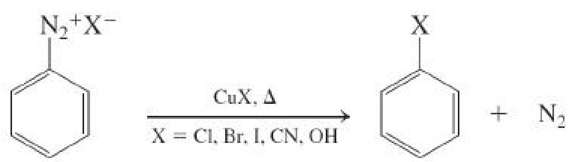
25. Vitamin C as an Antioxidant ([Section 22-9](#))



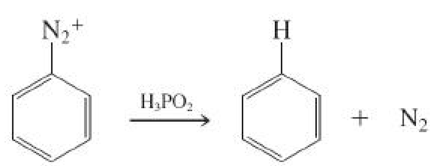
Semidehydroascorbic acid

Arenediazonium Salts

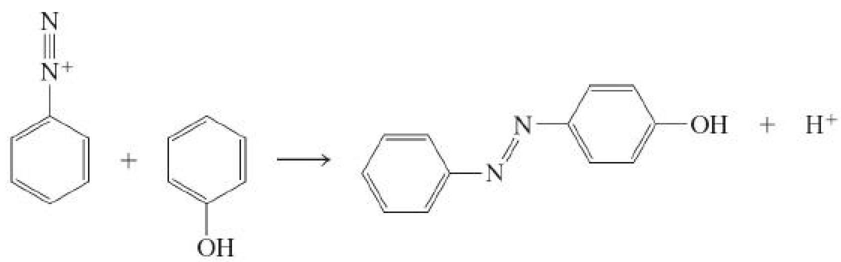
26. Sandmeyer Reactions ([Section 22-10](#))



27. Reduction ([Section 22-10](#))



28. Diazo Coupling ([Section 22-11](#))



Azo compound

Occurs only with strongly activated rings

Important Concepts

1. Phenylmethyl and other **benzylic radicals, cations,** and **anions** are reactive intermediates stabilized by **resonance** of the resulting centers with a benzene $\pi\pi$ system.
2. **Nucleophilic aromatic ipso substitution** accelerates with the nucleophilicity of the attacking species and with the number of electron-withdrawing groups on the ring, particularly if they are located ortho or para to the point of attack.
3. **Benzyne** is destabilized by the strain of the two distorted carbons forming the triple bond.
4. **Phenols** are aromatic enols, undergoing reactions typical of the hydroxy group and the aromatic ring.
5. **Benzoquinones** and benzenediols function as redox couples in the laboratory and in nature.
6. Vitamin E and the highly substituted phenol derivatives BHA and BHT function as inhibitors of the **radical-chain oxidation of lipids.** Vitamin C also is an antioxidant, capable of regenerating vitamin E at the surface of cell membranes.
7. **Arenediazonium ions** furnish reactive **aryl cations** whose positive charge cannot be delocalized into the aromatic ring.
8. The amino group can be used to direct electrophilic aromatic substitution, after which it is replaceable by diazotization and substitution, including reduction.